PulseTrack — Tier 1 Acceptance Checklist

What Tier 1 must present, and every check that has to pass before submission.

Derived from `candidate-brief.md` (the authority on scope), `questionnaire-dsma8.json`, `lab-results-template.csv` and the Definition of Done in `01-challenge-analysis.md` §19. Statuses reflect what has actually been run, not what is believed to work.

Part A — What Tier 1 is supposed to present

The brief is explicit that **"a complete, polished Tier 1 submission is a fully valid submission"** and that **"quality beats quantity every time."** Tier 1 is not a stepping stone to be rushed through — it is the thing being graded. Tier 2 adds depth on top of it, and an undocumented Tier 2 scores worse than a polished Tier 1.

The story the app has to tell

A small diabetes clinic. One clinician logs in and can run the whole loop without leaving the app:

1. See the state of the practice at a glance — how many patients, how many are answering their questionnaires, who is at risk.
2. Find or add a patient.
3. Email that patient a self-assessment link. The patient answers on their phone — **no account, no login** — and the score appears on their record.
4. Bulk-import lab results from a CSV the lab sent, including a messy one, and be told exactly what happened to every row.
5. Read the result as trends over time, per patient and across the clinic.

The six graded areas

AREA	WHAT "DONE" LOOKS LIKE
Authentication	Clinician email + password. **No patient accounts exist anywhere.** Be ready to justify the choice on the call.
Patient management	Full CRUD over name, DOB, sex, unique MRN, email, phone. List view with search. Real input validation.
Questionnaire flow	Tokenized email link → public form → 8 questions → server-computed score and band → history on the patient page. Links expire in 7 days and are single-use.
CSV lab upload	In-app template download, row-level validation, a per-row report saying *why*, partial import, and no duplicates on re-upload. **"We will test your uploader with a deliberately messy file. Ugly data is the point."**
Dashboards	Patient trends (at least glucose and HbA1c) plus score history; clinic aggregates, completion rate, per-band counts, recent uploads with a filter. **"A dashboard with no data should look intentional, not broken."**
Documentation	Setup under 10 minutes, architecture diagram, ERD, and Decisions & tradeoffs. **"We read this carefully."**

What the submission itself consists of

1. Git repository link

History intact and incremental — “we look at how you work, not just the end state.” Never squashed.

2. Live URL on Vercel

“Required, not optional.” Currently the only outstanding Tier 1 item.

3. A test clinician login

Sent in the submission email, not published in the public repo.

4. The README

Covering everything. Read carefully by the evaluators.

> **Then a walkthrough call.** You demo the app, then they ask detailed questions about the code and the decisions. Anything in the README’s “Decisions & tradeoffs” section is fair game — which is why each decision there states the cost it accepts, not just the choice.

Four things the brief signals it will actively probe

- **A deliberately messy CSV.** Named outright. Partial import and per-row reasons are the requirement, not graceful failure.
- **Security.** “This is healthcare.” Tokenized links that expire, no secrets in the repo, nothing sensitive in logs or URLs, parameterized queries, sensible authorization.
- **Empty, loading and error states.** Called out explicitly as detail they care about.
- **The README.** Stated twice: setup under 10 minutes, and decisions read carefully.

Part B — The acceptance checklist

AUTOMATED covered by `npm test` (188 tests)

VERIFIED confirmed against the running app, the live database, or the code itself

TO TEST not yet exercised — do before submitting BLOCKED needs the live Vercel URL

29	52	24	6
AUTOMATED	VERIFIED	TO TEST	BLOCKED

111 checks in total. The 24 marked *to test* and the 6 *blocked* are what stands between this and a submission that has actually been proven rather than assumed.

1 · Authentication

#	CHECK	HOW TO TEST / EXPECTED	STATUS
1.1	Valid credentials sign in	Log in as the seeded clinician → lands on `/dashboard`.	VERIFIED
1.2	Wrong password is rejected	Same email, wrong password → generic error. Must *not* say whether the account exists.	TO TEST
1.3	Unknown email is rejected identically	Same message and comparable response time as 1.2 — no user enumeration by text or timing.	TO TEST
1.4	Passwords stored as bcrypt hashes only	Inspect the `clinicians` row — `passwordHash` is a bcrypt string, no plaintext column exists.	VERIFIED
1.5	Protected page while signed out redirects	Sign out, open `/patients` directly → redirected to `/login`.	TO TEST
1.6	API route while signed out returns 401 JSON	`curl -X POST /api/labs/upload` with no cookie → **401 JSON**, never a 307 to `/login`.	VERIFIED
1.7	Server action rejects an unauthenticated caller	Every mutation calls `requireClinician()`; hidden UI is not the boundary.	VERIFIED
1.8	Sign out ends the session	Sign out, press Back → the protected page does not render from cache.	TO TEST
1.9	No patient can authenticate anywhere	No patient account model exists; patients reach the form by token only.	VERIFIED

2 · Patient management

#	CHECK	HOW TO TEST / EXPECTED	STATUS
2.1	Create a patient	All six fields save and appear in the list.	VERIFIED

#	CHECK	HOW TO TEST / EXPECTED	STATUS
2.2	Read / detail page	Opens with details, charts, assessments and labs.	VERIFIED
2.3	Update a patient	Edit, save, value persists after reload.	TO TEST
2.4	Delete a patient	Confirm step, then the patient and their assessments and labs are gone (cascade).	TO TEST
2.5	Duplicate MRN rejected	Create a second patient with an existing MRN → friendly error, not a 500. Enforced by a DB unique index.	VERIFIED
2.6	Search by name	Partial, case-insensitive match returns the patient.	VERIFIED
2.7	Search by MRN	Returns the exact patient.	TO TEST
2.8	Search with no matches	Shows a designed empty state, not a blank page.	TO TEST
2.9	Required fields validated	Submit empty → per-field messages.	AUTOMATED
2.10	Future date of birth rejected	Tomorrow's date → "Date of birth cannot be in the future."	AUTOMATED
2.11	Malformed email rejected	`not-an-email` → clear message.	AUTOMATED
2.12	Sex accepts only the enum	Values map to FHIR `Patient.gender`.	AUTOMATED
2.13	DOB renders on the correct calendar day	Stored `@db.Date`; must not shift a day west of UTC.	AUTOMATED

3 · Assessment flow

#	CHECK	HOW TO TEST / EXPECTED	STATUS
3.1	Send assessment emails a tokenized link	Click Send → email dispatched (console adapter in dev) and a copy-link shown in the UI.	VERIFIED
3.2	Sending requires a patient email	Patient with no email → clear message, not a crash.	TO TEST
3.3	Token is ≥32 random bytes	Generated with a CSPRNG.	AUTOMATED
3.4	Only the SHA-256 hash is stored	The `assessments` row holds `tokenHash`; the raw token appears nowhere in the DB.	AUTOMATED
3.5	Link works with no login	Open in a fresh cookie-less browser context → the form renders.	VERIFIED
3.6	All 8 questions and 4 options render exactly as the official JSON	8 fieldsets, 32 radios, wording identical to `questionnaire-dsma8.json`.	VERIFIED

#	CHECK	HOW TO TEST / EXPECTED	STATUS
3.7	Incomplete submission is blocked	Answer 7 of 8 → refused with a clear message (`allItemsRequired: true`).	AUTOMATED
3.8	Score computed server-side	Sum of the 8 answers, recomputed from stored answers.	AUTOMATED
3.9	A client-supplied score is ignored	POST a forged `totalScore` → the stored score still equals the sum of answers.	VERIFIED
3.10	Band boundaries, all eight	0 and 6 → Low · 7 and 12 → Moderate · 13 and 18 → High · 19 and 24 → Very high.	AUTOMATED
3.11	Link is single-use	Submit, then reopen the same link → "already used" screen, not a second submission.	AUTOMATED
3.12	Concurrent double-submit writes one completion	Two simultaneous POSTs → 1 completion and 8 answers, not 16. Enforced by a unique index.	VERIFIED
3.13	Link expires exactly 7 days after sending	`expiresAt = sentAt + 7d` ; an expired token is refused.	AUTOMATED
3.14	Invalid, expired and used links each get a distinct screen	Three different messages, none revealing whether the token ever existed.	TO TEST
3.15	Status tracked `sent → completed | expired`	Visible per assessment on the patient page.	VERIFIED
3.16	Assessment history shows on the patient page	Sent date, status, completed date, score /24, band.	VERIFIED
3.17	The public form is usable on a phone	375px: full-width tap targets, no sideways scroll.	VERIFIED

4 · CSV lab import — the area the brief says it will attack

#	CHECK	HOW TO TEST / EXPECTED	STATUS
4.1	Template downloads from inside the app	"Download template" returns a CSV **byte-identical** to the supplied `lab-results-template.csv`.	VERIFIED
4.2	The supplied clean sample imports 10/10	On a fresh database, all ten rows accepted.	VERIFIED
4.3	Re-uploading the same file creates zero duplicates	Second upload: 0 accepted, 0 rejected, 10 "already imported". Row count stays 10.	VERIFIED
4.4	Partial import — valid rows land even when others fail	Messy file imports its good rows and explains every other outcome.	VERIFIED
4.5	Per-row report says *why*	Every rejected row names its reason and its real file line number.	VERIFIED
4.6	Unknown MRN rejected	Row naming a patient that does not exist.	AUTOMATED
4.7	Malformed date rejected	`2026-13-45`, `not-a-date`.	AUTOMATED
4.8	Future collected date rejected	Explicitly named in the brief.	AUTOMATED
4.9	Non-numeric value rejected	`abc`, empty, `--`.	AUTOMATED
4.10	Missing required field rejected	Required = `mrn`, `collected_date`, `test_code`, `value`.	AUTOMATED
4.11	Unknown test code rejected	Anything outside `GLU-F`, `HBA1C`, `SBP`.	AUTOMATED
4.12	Duplicate within one file	Same MRN + date + test twice in the same upload → one import, one reported duplicate.	AUTOMATED
4.13	Duplicate against rows already on file	Reported as "already imported", which is **not** an error.	AUTOMATED
4.14	Same key, different value → skipped and reported	Never silently overwrites a stored measurement (D-CSV-2).	AUTOMATED
4.15	BOM, CRLF, quoted fields, ragged rows all parse	Including a **short first data row**, which previously rejected whole files.	AUTOMATED
4.16	Cosmetic mismatch warns, does not reject	Different `test_name` or a broken reference range imports with a flag.	AUTOMATED
4.17	Mismatched unit stored as reported, never relabelled	`5.8 mmol/L` stays `mmol/L` and is flagged (D-CSV-5).	AUTOMATED
4.18	Wrong file type fails cleanly	`.txt` upload → 415, a readable message, no stack trace.	VERIFIED
4.19	Wrong columns fail cleanly	CSV missing required headers → 422 naming what is missing.	VERIFIED

#	CHECK	HOW TO TEST / EXPECTED	STATUS
4.20	Empty file / header-only file	Told there are no data rows; not a crash and not a silent success.	AUTOMATED
4.21	Oversized file refused	Above the 2 MB limit → refused with a clear message.	TO TEST
4.22	Import appears immediately on the patient's charts	Upload, open the patient → the new points are plotted.	TO TEST

> **Build one deliberately ugly file before submitting** and keep it in the repo. The brief says it will test with one. It should contain, in a single file: a BOM, CRLF line endings, a short first data row, a quoted field containing a comma, a blank line mid-file, an unknown MRN, a future date, a malformed date, a non-numeric value, an unknown test code, a within-file duplicate, a row already on file, a mismatched unit, and an inverted reference range. Then check the report explains all of them and still imports the valid rows.

5 · Dashboards — patient view

#	CHECK	HOW TO TEST / EXPECTED	STATUS
5.1	Glucose trend over time	Required by the brief by name.	VERIFIED
5.2	HbA1c trend over time	Required by the brief by name.	VERIFIED
5.3	Questionnaire score history	DSMA-8 total over time, 0–24 axis.	VERIFIED
5.4	Chronology is correct	Numeric time axis: a back-dated result must not double the line back on itself, and a year's gap must not look like a day's.	AUTOMATED
5.5	A single reading still renders	Zero-width domain is padded so the marker does not vanish. Every brand-new patient hits this.	AUTOMATED
5.6	Units and reference ranges shown	mg/dL, %, mmHg with the range stated.	VERIFIED
5.7	Tooltips work on hover	Brief asks for "proper time-series charts with sensible axes".	TO TEST
5.8	A patient with no data looks intentional	Create a fresh patient → designed empty states, not four broken boxes.	TO TEST
5.9	Values are reachable without hovering	A lab table accompanies the charts, for keyboard and touch users.	VERIFIED

6 · Dashboards — clinic view

#	CHECK	HOW TO TEST / EXPECTED	STATUS
6.1	Total patients	Matches `SELECT count(*) FROM patients`.	VERIFIED
6.2	Assessment completion rate	completed ÷ all sent, all time. Currently 88% = 7 of 8.	VERIFIED

#	CHECK	HOW TO TEST / EXPECTED	STATUS
6.3	Completion rate with a zero denominator	Shows `–`, never `0%`, when nothing has been sent.	AUTOMATED
6.4	Patients per risk band	Each patient counted **once**, by their latest completed assessment.	VERIFIED
6.5	Band counts sum to the patient total	Including a "No assessment" bucket.	VERIFIED
6.6	Aggregates verified by hand against the data	Named in the Definition of Done. Jane has a High-risk assessment in her history yet High reads 0 — proof it counts patients, not assessments.	VERIFIED
6.7	Recent uploads panel	Filename, imported / rejected / already-on-file counts, timestamp.	VERIFIED
6.8	At least one working filter	Date range: 7 / 30 / 90 days / all time actually changes the rows.	TO TEST
6.9	Risk bands distinguishable without colour	Every band carries its own label and number; no two fills touch.	VERIFIED

7 · States, responsiveness and accessibility

#	CHECK	HOW TO TEST / EXPECTED	STATUS
7.1	Empty database looks intentional	Called out in the brief. Test on a database with zero patients — the hardest state to get right.	TO TEST
7.2	Loading states on every async panel	Throttle the network and watch each panel: skeletons, not layout jumps.	TO TEST
7.3	Error states	Stop the database and load a page → a readable error, not a stack trace.	TO TEST
7.4	Usable at 375px	Zero horizontal overflow on every page. Verified in Chrome across six pages at 375 and 1280.	VERIFIED
7.5	No text truncated by a fixed-width container	Check at 1280px too — a `truncate` hides text at every width, not just mobile.	VERIFIED
7.6	Charts render in a real browser	Recharts draws client-side; SSR and jsdom cannot verify this. Headless Chrome can.	VERIFIED
7.7	Dates render on the correct calendar day everywhere	List, detail, charts, CSV report, upload timestamps.	AUTOMATED
7.8	Keyboard navigation through a form	Tab order sensible; focus visible; the questionnaire completable without a mouse.	TO TEST
7.9	404 for a patient id that does not exist	`/patients/does-not-exist` → a designed not-found page.	TO TEST

8 · Security and privacy

#	CHECK	HOW TO TEST / EXPECTED	STATUS
8.1	No secret anywhere in git history	`git log --all -p | grep <value>` for every real value in `.env`.	VERIFIED
8.2	No secret in the client bundle	Nothing sensitive behind `NEXT_PUBLIC_`; grep the built JS for the API key.	TO TEST
8.3	No patient data in logs	Ids and counts only — never names, emails, MRNs, answers or tokens.	VERIFIED
8.4	No sensitive data in URLs	The assessment token is the deliberate exception and is the access grant itself.	VERIFIED
8.5	Parameterized queries throughout	Prisma everywhere; no string-built SQL.	VERIFIED
8.6	Authorization checked server-side on every mutation	Not just hidden UI and not just the edge proxy.	VERIFIED
8.7	Only fabricated data	No real patient anywhere, including in fixtures and screenshots.	VERIFIED

#	CHECK	HOW TO TEST / EXPECTED	STATUS
8.8	`.env` is gitignored and never committed	Confirm `git ls-files .env` is empty.	VERIFIED
8.9	No HIPAA or GDPR compliance claimed	Good practice is not compliance; the README says so explicitly.	VERIFIED

9 · Documentation and repository

#	CHECK	HOW TO TEST / EXPECTED	STATUS
9.1	Setup runnable in under 10 minutes	**Clone into a clean directory and follow the README literally** — the only honest test.	TO TEST
9.2	Architecture diagram present and renders on GitHub	Mermaid — check it renders on the GitHub page, not just locally.	TO TEST
9.3	ERD present and renders on GitHub	Same check.	TO TEST
9.4	"Decisions & tradeoffs" section	15 numbered decisions, each stating what it gives up.	VERIFIED
9.5	Git history incremental and meaningful	14 PRs, merge commits, never squashed, branches kept.	VERIFIED
9.6	Test suite green	`npm test` → 188 passing.	VERIFIED
9.7	Lint clean	`npm run lint` → no output.	VERIFIED
9.8	Production build succeeds	`npm run build`.	VERIFIED

10 · Deployment — blocked on the Vercel account

#	CHECK	HOW TO TEST / EXPECTED	STATUS
10.1	App is live on a Vercel URL	"Required, not optional."	BLOCKED
10.2	Migrations run against the production database	Build command `prisma migrate deploy && next build`.	BLOCKED
10.3	Seed run in production so the evaluator's login exists	Definition of Done: "the evaluator's test account is valid."	BLOCKED
10.4	`APP_BASE_URL` set to the deployed origin	Otherwise emailed assessment links point at `localhost`.	BLOCKED
10.5	No Postgres connection exhaustion under clicking	Pooled URL at runtime; watch for intermittent 500s.	BLOCKED
10.6	Re-run this entire checklist against the live URL	Local passing does not imply deployed passing.	BLOCKED

> **The highest-value untested item is 9.1.** Everything else is a feature check; 9.1 is the one an evaluator performs *first*, on a machine that is not yours, and a README that only works because your environment already has something set up will fail in front of them. Clone into an empty directory with a fresh database and follow your own instructions literally.

> **Two things to remember while testing.** Sending an assessment adds a row and moves the headline completion rate off the documented 88% — delete it afterwards. And the demo database should end as: 3 patients, 8 assessments (7 completed, 1 expired), 10 lab results.

PulseTrack · Tier 1 acceptance checklist · generated 2026-08-24 · sources: `candidate-brief.md`, `questionnaire-dsma8.json`, `lab-results-template.csv`, `01-challenge-analysis.md` §19